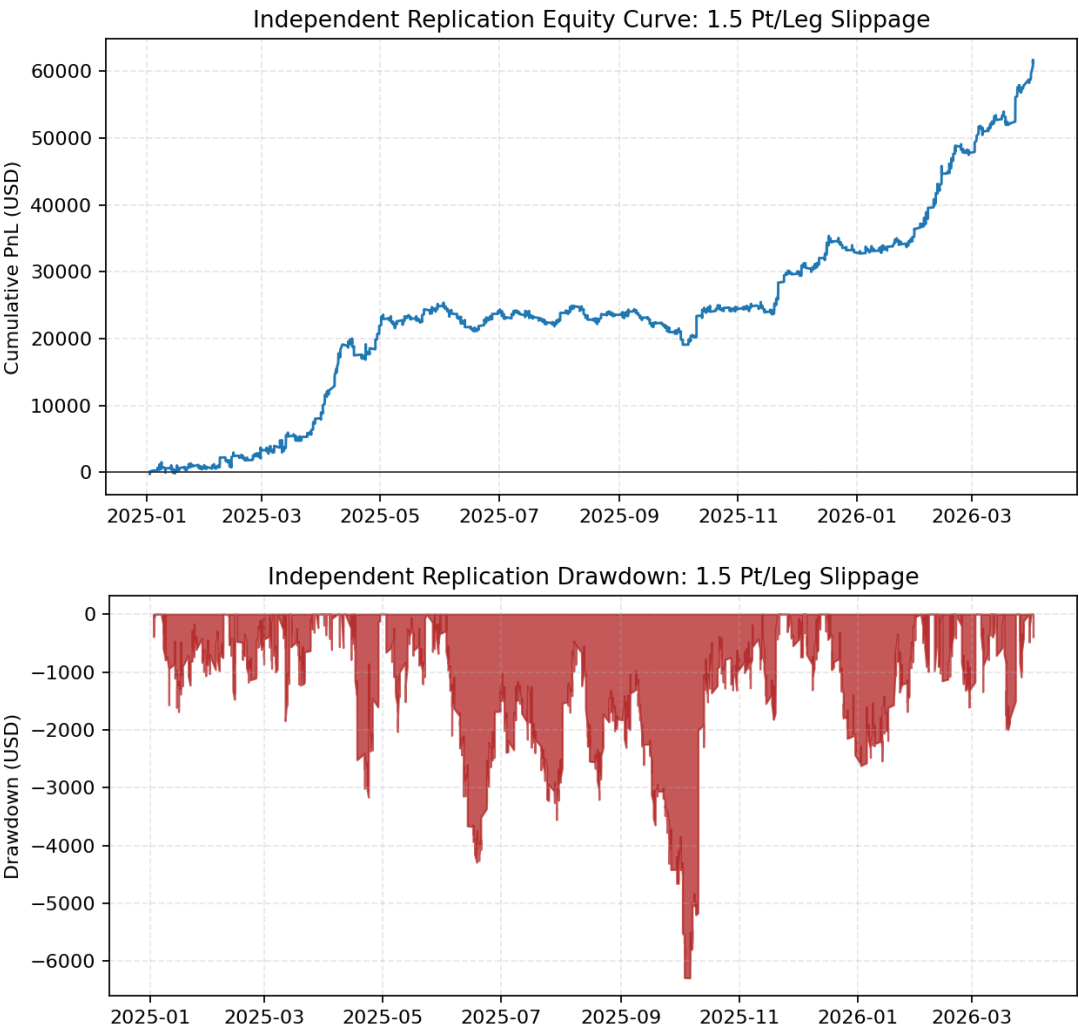


NQ Futures Strategy Overfitting and Replication Audit

Independent rebuild, leakage checks, stress tests, and claim reconciliation

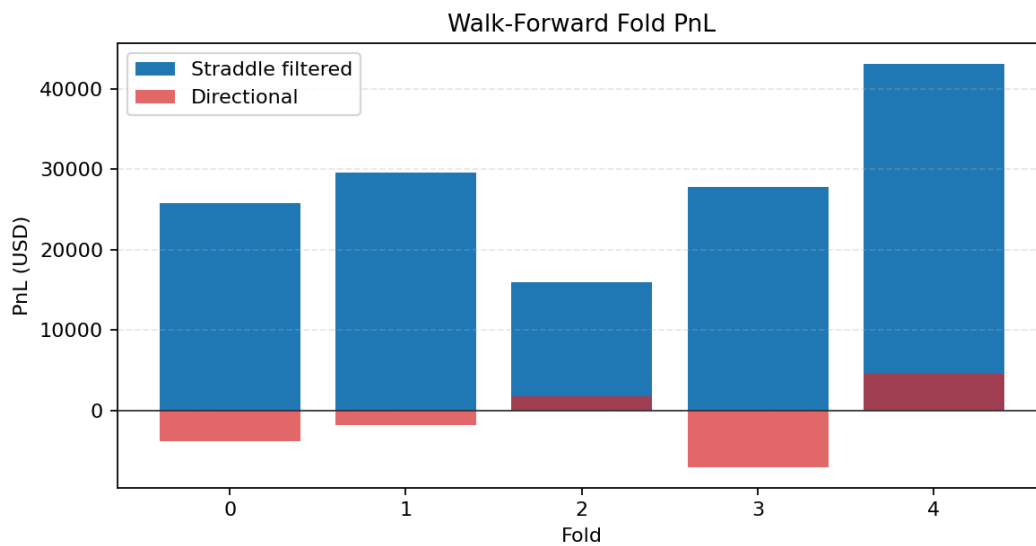
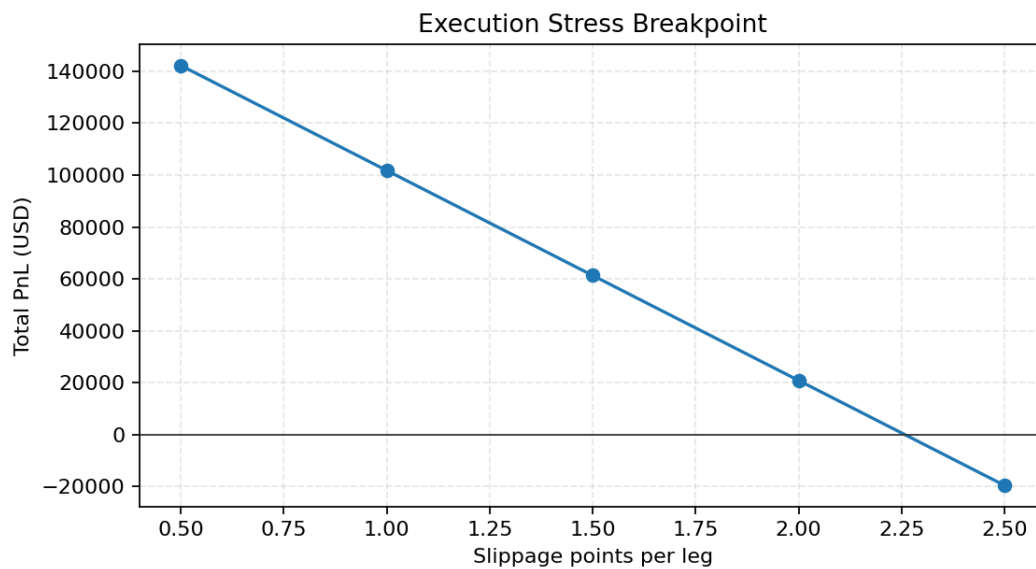
This audit freezes the existing strategy instead of tuning for a better result. The original conservative result was independently rebuilt from raw NQ bars, then retested under cleaner validation and execution stress.

Test	Trades	PnL	Win	PF	Monthly Sharpe
Replicated original 1.5 pt	2,022	\$61,301	66.6%	1.38	2.85
Clean eval 1.5 pt	1,925	\$60,511	67.1%	1.39	2.91
Purged clean eval 1.5 pt	1,923	\$56,145	66.4%	1.36	2.87
Label-shuffle null 1.5 pt	0	\$0	0.0%	nan	nan



Key correction: the old 9+ or 13+ value is a trade-level t-statistic, not a finance-style Sharpe ratio.

Validation caveat: the original script used the test fold for early stopping; cleaner train-tail validation is reported separately.



Claim verdict counts

Verdict	Count
supported	19
rejected	1
corrected	1

Failure checks

Failure Check	Result	Meaning
Same-bar adverse order	\$-175,264	Execution assumption can kill result
Cross ES	\$-11,847	No broad futures generalization
Cross MGC	\$-9,625	No broad futures generalization

Leakage and validation checks

Check	Verdict	Short evidence
future_shift_in_features	pass	No negative shift found in feature assignments.

Check	Verdict	Short evidence
future_shift_in_labels	allowed	Negative shift appears in labels only: 103: f["lv"] = ((h.shift(-1) - l.shift(-1)) > 1.2 ...
original_fold_boundary_overlap	warn	Original pandas slices share 5 timestamp rows across train/test boundaries.
test_fold_used_for_early_stopping	warn	Original LightGBM fit uses eval_set=[(Xte, te[label])] for early stopping; audit adds tra...
old_monte_carlo_meaning	warn	Original Monte Carlo permutes final trade PnL, preserving total PnL; it is a path-order d...

Block bootstrap

Block	P(pos)	P05 PnL	Median PnL	Worst PnL	Worst DD	Sharpe P05/Med
day	100.0%	\$41,842	\$60,985	\$20,506	\$-9,897	3.31/4.58
month	100.0%	\$32,791	\$60,105	\$4,435	\$-13,782	1.85/2.94
week	100.0%	\$36,590	\$61,211	\$10,518	\$-12,769	2.30/3.50

